

Table 7-1 shows the typical start-up time for a number of standard output voltages.

Table 7-1. Typical Start-Up Times

OUTPUT VOLTAGE	SOFT-START RAMP-UP TIME ($t_{r(SS)}$)	START-UP TIME ($t_{d(EN)} + t_{r(SS)}$)
1.8 V	0.5 ms	2 ms
2.5 V	1.375 ms	2.875 ms
3.3 V	2.375 ms	3.875 ms
5 V	4.5 ms	6 ms

If the output is prebiased – that is, the initial output voltage is not zero – the start-up behavior is as follows:

- If the prebias voltage is *lower* than the target voltage, the device does not start switching until the ramping output voltage is greater than the prebias voltage (see Figure 7-8).
- If the prebias voltage is *higher* than the target voltage, the device does not start to switch until the output voltage has decreased to the target voltage (see Figure 7-9). The device cannot actively discharge the output to the target voltage and relies on the load current to discharge the output capacitor and decrease the output voltage to the target value.

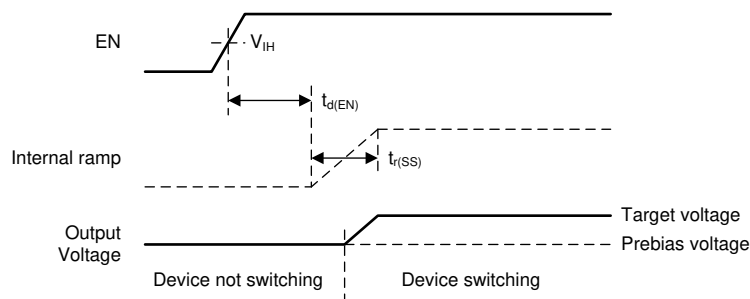


Figure 7-8. Start-Up Behavior into Prebiased (Low) Output

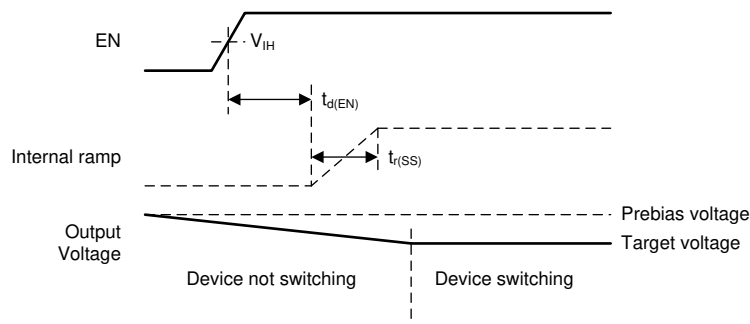


Figure 7-9. Start-Up Behavior into Prebiased (High) Output

7.3.4 Input Current Limit

The device can limit the current drawn from its supply, so that it can be used with batteries that do not support high peak currents. The input current limit is active during normal operation and at start-up to avoid high inrush current. The device has eight current limit settings:

- 1 mA
- 2.5 mA
- 5 mA
- 10 mA
- 25 mA
- 50 mA

- 100 mA
- Unlimited

CFG1 and CFG2 pins select which setting is active (see [Section 7.3.6](#)).

7.3.5 Dynamic Voltage Scaling

The device has a dynamic voltage scaling function to switch between the two output voltage settings. When the SEL pin changes state, the output voltage ramps to the new value in 100-mV steps. The duration of each step is 125 μ s (see [Figure 7-10](#)).

The device does not actively discharge the output capacitor, when the output voltage ramps to a lower level. This leads to a longer output voltage settling time when light load is applied (see [Figure 7-11](#)). The settling time can be calculated with [Equation 3](#).

$$t_{\text{settle}} = C_O \times \frac{V_{O(\text{HIGH})} - V_{O(\text{LOW})}}{I_O} \quad (3)$$

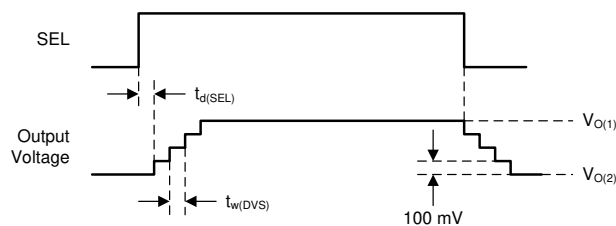


Figure 7-10. Dynamic Voltage Scaling with High Load

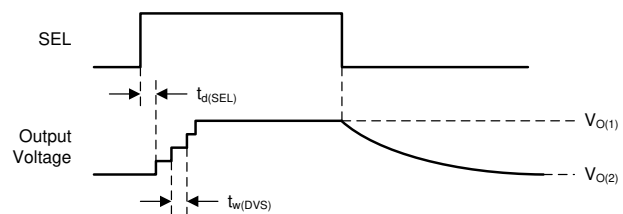


Figure 7-11. Dynamic Voltage Scaling with Light Load

7.3.6 Device Configuration (Resistor-to-Digital Interface)

The device has three configuration pins (CFG1, CFG2, and CFG3) that control its operation. When the device starts up, a resistor-to-digital (R2D) interface reads the values of the configuration resistors on the CFG pins and transfers the setting to an internal configuration register (see [Figure 7-12](#)).

- CFG1 and CFG2 set $V_{O(2)}$ level and the input current limit.
- CFG3 sets $V_{O(1)}$ level.

To reduce power consumption, the device reads the value of the resistors connected to the configuration pins during start-up and then disables these pins. Once the device has started to operate, changes to the configuration pins have no effect.

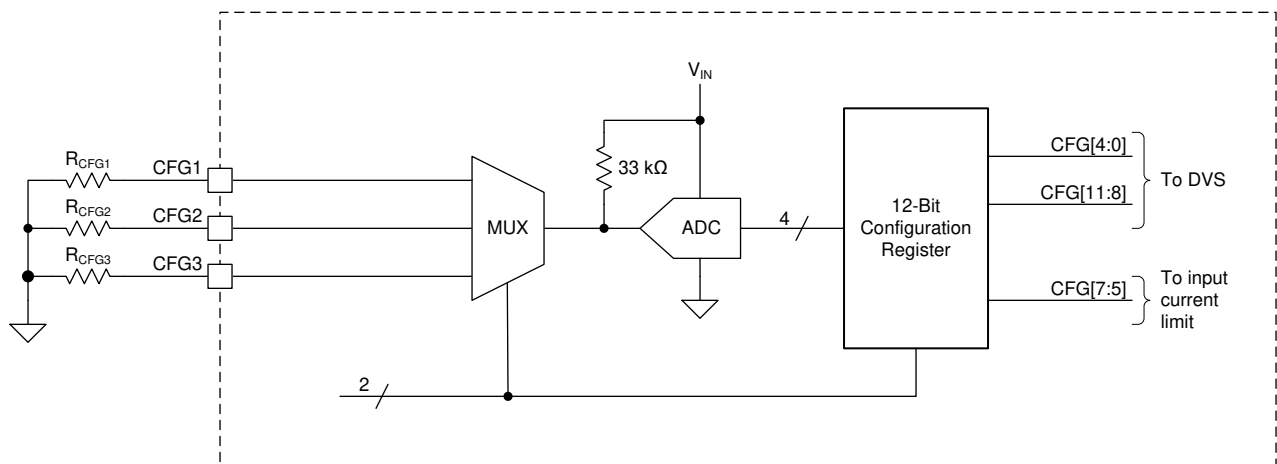


Figure 7-12. Resistor-to-Digital Interface Block Diagram